The Impact of Substances on Sleep Quality

Bradley, Siera, Haydn, Mamadou



TABLE OF CONTENTS

01

Our Data and Variables

02

Impacts on Sleep Efficiency

03

Impacts on Deep Sleep

04

Conclusion

01

DATA AND VARIABLES

Our Dataset

Sleep Efficiency - Kaggle
452 Rows
15 Columns

Information on test subjects and their sleep patterns.

Chosen Variables

Deep sleep percentage
REM sleep percentage
Sleep efficiency
Caffeine consumption
Awakenings
Exercise frequency
Alcohol consumption
Gender
Smoking status

Current study

Our project explores the effects of alcohol and caffeine use on sleep efficiency and deep sleep. Sleep efficiency and deep sleep have long been used by researchers to measure sleep quality and sleep disorders (Reed & Sacco, 2016; Riedel & Lichstein, 1998). Our analysis revealed substance use has similar and different effects for males and females, with stronger effects for those who smoke compared to those who do not smoke.

Importance of sleep

- Getting adequate sleep is important for cognition, mood, mental and physical health (Watson et al., 2015).
- Even short-term sleep deprivation can have a detrimental effect on physical and mental health, as well as public safety (Luyster et al., 2012).

Substance use

- About 85% of the U.S. population consumes at least one caffeinated beverage per day, containing about 165 mg. of caffeine (Mitchell et al., 2014).
- As of February, 2025, about 79% of the U.S. population ages 12 and older reported they drank alcohol at some point in their lifetime (National Institute on Alcohol Abuse and Alcoholism, 2025)

IMPORTANT SLEEP TERMS

Sleep Quality

- is evaluated on multiple parameters such as awake time during the night, number of awakenings, total sleep time or amount of deep sleep.

Sleep Efficiency

- is the ratio of time one spends in bed and sleeping.

Deep Sleep

- is a stage of sleep characterized by slow brain waves and a deep state of relaxation where the body repairs and regenerates tissues, builds bone and muscle, and strengthens the immune system.

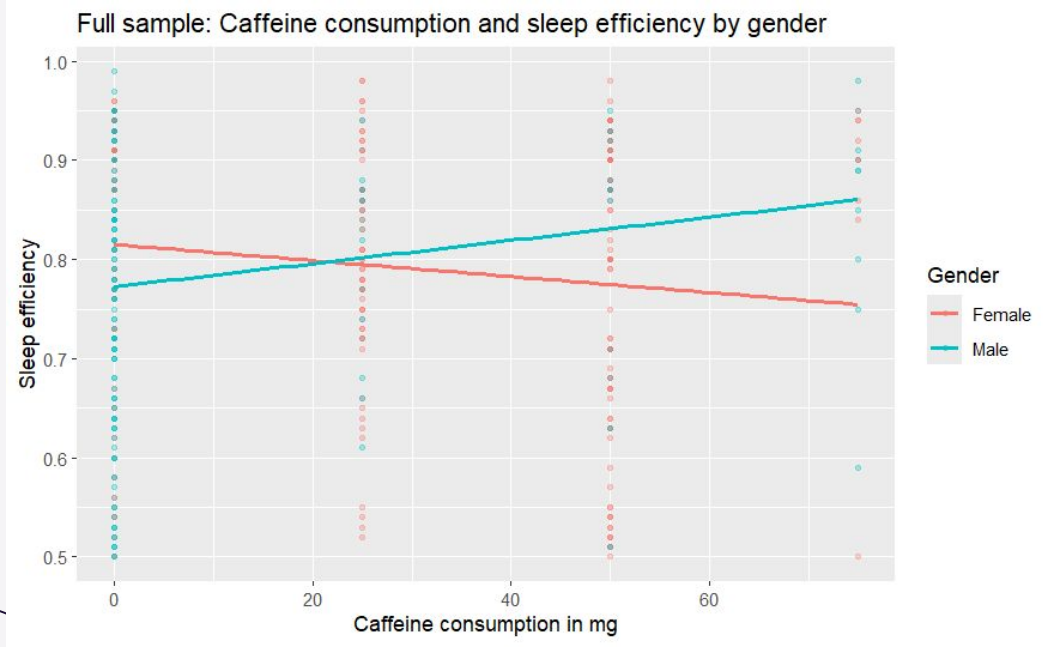
REM Sleep

- is a stage of sleep characterized by rapid eye movements, increased brain activity, and temporary muscle paralysis. During REM sleep, the brain is highly active, similar to when you are awake, and is when most dreams occur.

02

IMPACTS ON SLEEP EFFICIENCY

Caffeine Impacts on Sleep Efficiency



Findings

Caffeine consumption is shown to affect the sleep efficiency of men and women differently.

For women: one additional mg of caffeine consumed is associated with a **decrease** in sleep efficiency of **0.0008 (one soda, 40 mg = 0.033)**

For men: one additional mg of caffeine consumed is associated with an **increase** in sleep efficiency of **0.0012 (one soda, 40mg = 0.048)**

More than a quarter (28%) of men work overtime everyday compared to 13% of women (Office Freedom, 2022)

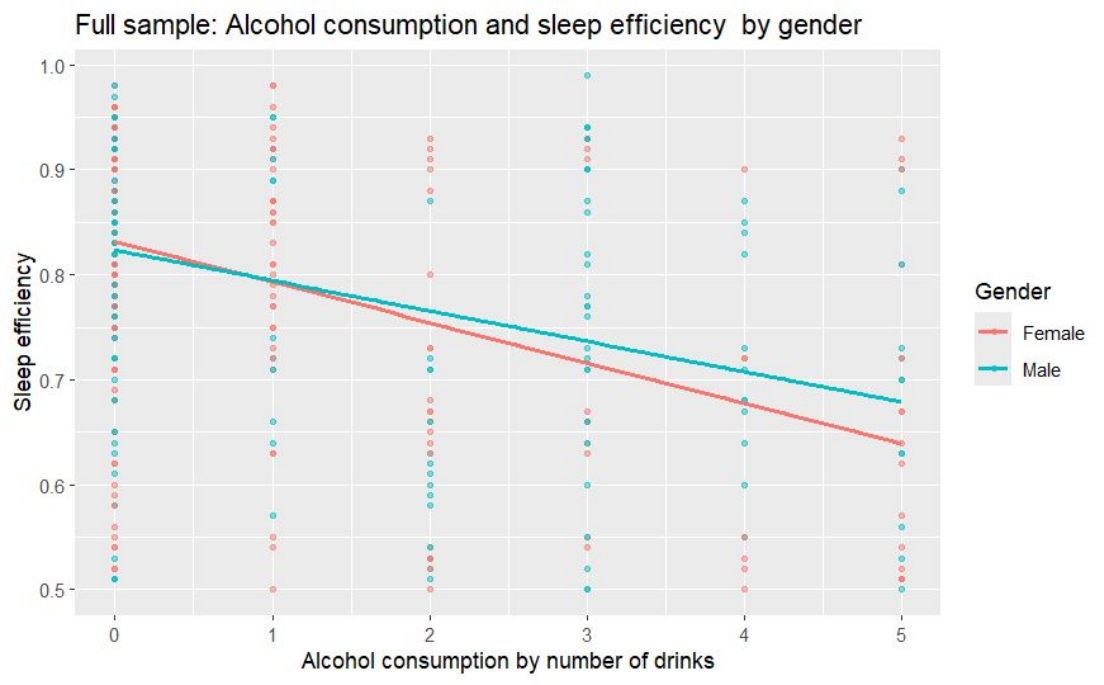
Alcohol Impacts on Sleep Efficiency

Findings

Increased alcohol consumption has a negative impact on Sleep Efficiency

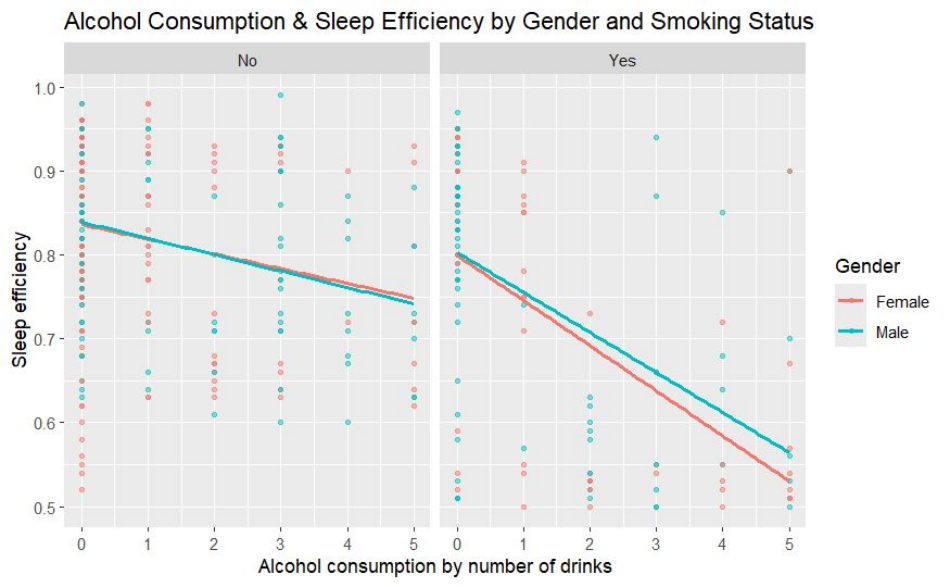
For women: one additional drink consumed is associated with a **decrease** in sleep efficiency of **0.038**

For men: one additional drink consumed is associated with a **decrease** in sleep efficiency of **0.029**



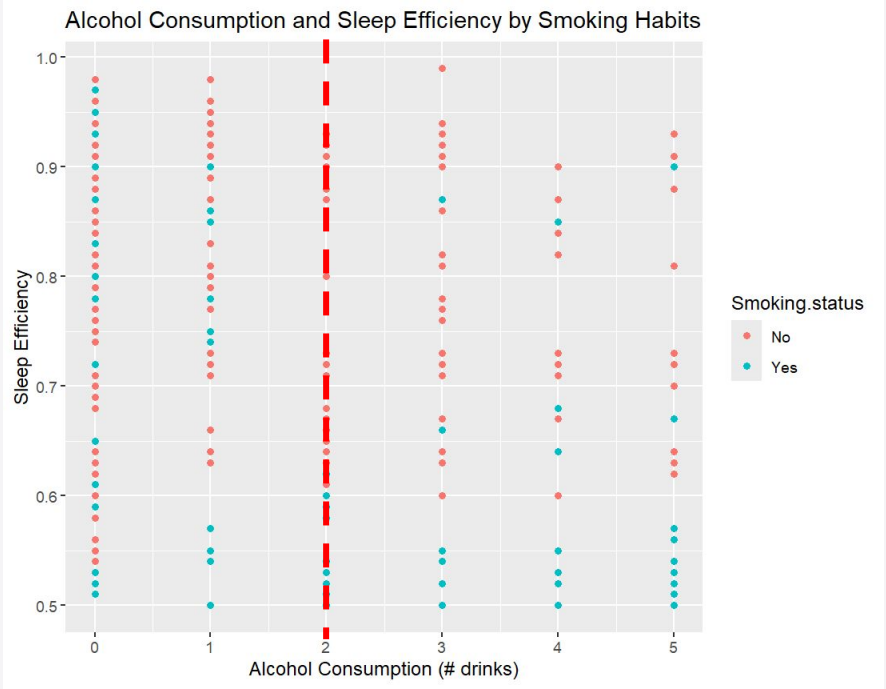
The enzyme that begins the metabolization of alcohol in the stomach, is found in significantly higher concentrations in men (about 50% more) than in women (Alcohol and Drug Program Student Health Service, 2021)

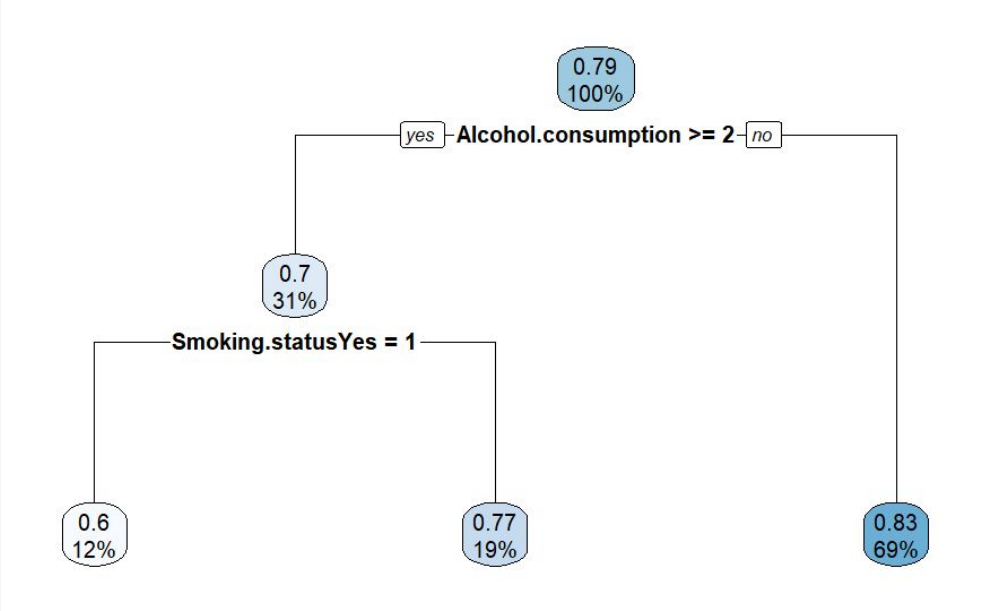
Alcohol Impact on Sleep Efficiency by Smoking habits



Steeper negative slope for smokers' impact of alcohol consumption on sleep efficiency (and impacts women more)

Prediction of Sleep Efficiency



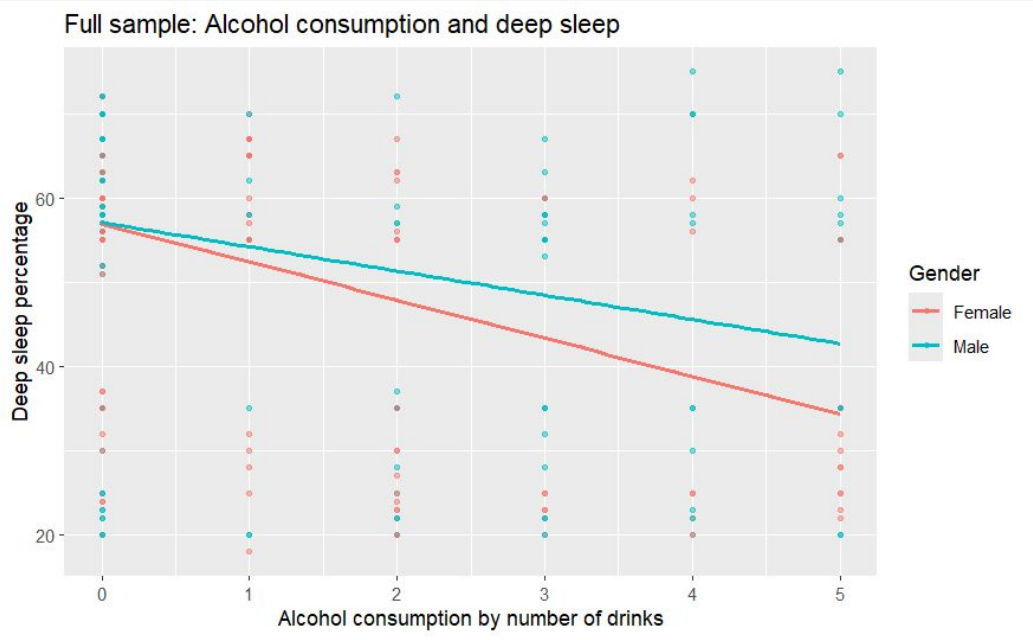


Using smoking habits, alcohol consumption, caffeine consumption, and gender

03

IMPACTS ON DEEP SLEEP

Alcohol Consumption Impacts on Deep



Findings

Increased alcohol consumption has a negative impact on Sleep Efficiency

For women: one additional drink consumed is associated with a decrease in sleep efficiency of **-4.526928**

For men: one additional drink consumed is associated with a decrease in sleep efficiency of **-2.874828**

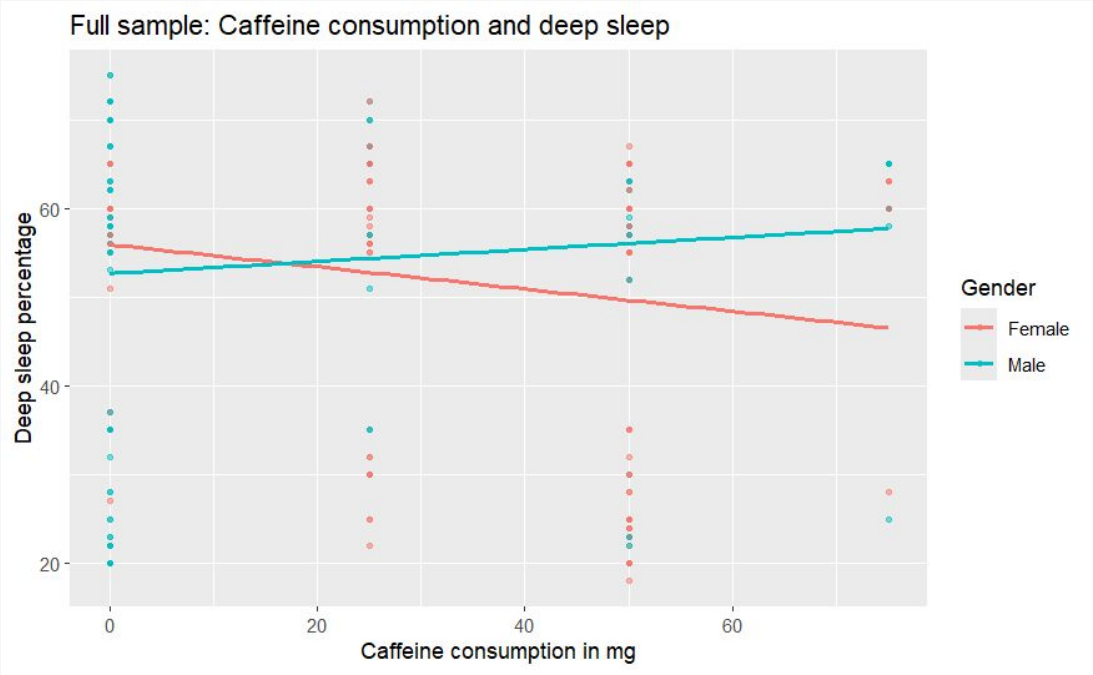
Caffeine Impacts on Deep Sleep

Findings

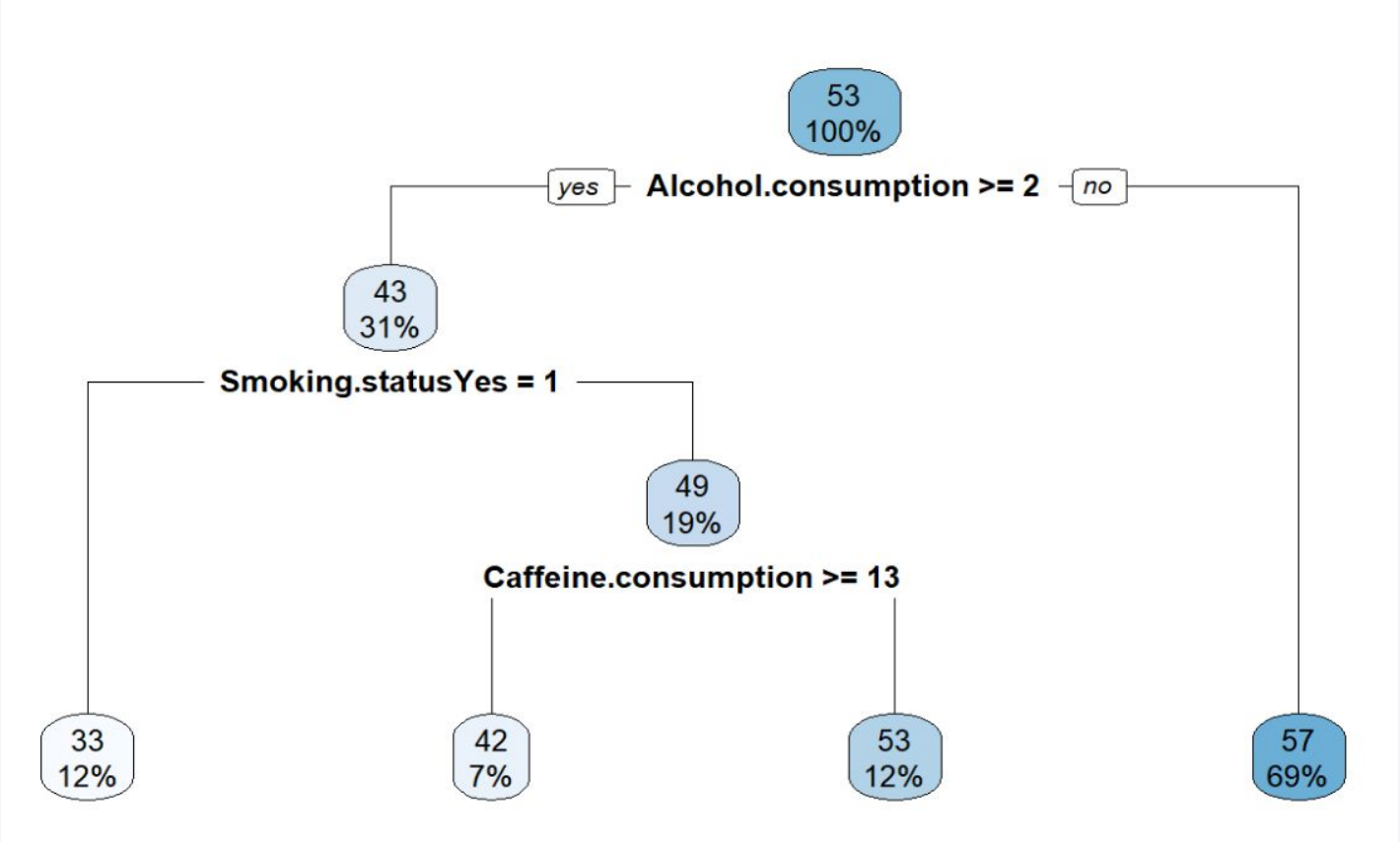
Similar to sleep efficiency caffeine consumption is shown to affect the deep sleep percentage of men and women differently.

For women: one additional mg of caffeine consumed is associated with a decrease in sleep efficiency of about **-0.1252147**

For men: one additional mg of caffeine consumed is associated with an increase in sleep efficiency of **0.0503813**



Prediction of Deep Sleep



04

CONCLUSIONS

Conclusions



Impacts of Caffeine

Caffeine had a positive effect on males and not super steep negative effect on females. This makes our findings inconclusive.



Impacts of Alcohol

For both males and females, the use of alcohol is negatively associated with sleep quality. We see a 19% decrease for females and 16% for males as the drinks go from 0-5.



Impacts of Smoking

We saw a 30 % decrease in sleep efficiency for smokers compared to non-smokers. This suggests that some interaction between nicotine and alcohol takes place but not enough to make a conclusion.



Future Work

This dataset is purely observational. If we were to do a future study we would like to incorporate our own parameters to better understand effects on sleeping.

Sources

Alcohol and Drug Program Student Health Service. (2021, May). *Alcohol 101: Gender differences*. University of California, Santa Barbara. https://adp.sa.ucsb.edu/sites/default/files/2021-05/alcohol_-_genderdifferences.pdf

Luyster, F. S., Strollo, P. J., Jr., Zee, P. C., & Walsh, J. K. (2012). Sleep: A health imperative. *Sleep*, *35*(6), 727–734. https://doi.org/10.5665/sleep.1846

Mitchell, D. C., Knight, C. A., Hockenberry, J., Teplansky, R., & Hartman, T. J. (2014). Beverage caffeine intakes in the U.S. *Food and Chemical Toxicology: An International Journal Published for the British Industrial Biological Research Association*, *63*, 136–142. https://doi.org/10.1016/j.fct.2013.10.042

National Institute on Alcohol Abuse and Alcoholism. (2025, February). *Alcohol Use in the United States: Age Groups and Demographic Characteristics*. https://www.niaaa.nih.gov/alcohols-effects-health/alcohol-topics-z/alcohol-facts-and-statistics/alcohol-use-united-states-age-groups-and-demographic-characteristics

Omoigui, N. (2022, June 23). *Men working more overtime than women*. HR Magazine. https://www.hrmagazine.co.uk/content/news/men-working-more-overtime-than-women/

Watson, N. F., Badr, M. S., Belenky, G., Bliwise, D. L., Buxton, O. M., Buysse, D., Dinges, D. F., Gangwisch, J., Grandner, M. A., Kushida, C., Malhotra, R. K., Martin, J. L., Patel, S. R., Quan, S. F., & Tasali, E. (2015). Joint consensus statement of the American Academy of Sleep Medicine and Sleep Research Society on the recommended amount of sleep for a healthy adult: Methodology and discussion. *Journal of Clinical Sleep Medicine*. https://doi.org/10.5664/jcsm.4950